Second Partial

Fco. Vazquez

Tec de Monterrey

Derivative of Exponential functions $f(x) = n^x$

We know how to compute the derivative of x^n using the power rule: $\frac{\mathrm{d}}{\mathrm{d}x}x^n = nx^{n-1}$. What if we change the symbols? $x^n \to n^x$. How we can compute the derivative of an exponential function?

Hint: Express the exponential function in terms of e^x

We know that $\frac{\mathrm{d}}{\mathrm{d}x}e^x = e^x$. Therefore, the big idea is to express the function n^x in terms of e^x.

Example: $f(x) = 2^x$

Express 2^x in terms of e^x

We can use the propoerty of inverse functions: $e^{\ln(x)} = x$. But instead of using x, let's use 2^x,

$$e^{\ln(2^x)} = 2^x.$$

Now, by using properties of logarithms, $\ln(x^n) = n\ln(x)$,

$$e^{x\ln(2)} = \left[e^{\ln(2)}\right]^x = 2^x.$$

Example: $f(x) = 2^x$

Use the chain rule to compute the derivative

Now, we can compute the derivative of $f(x)$

$$\begin{aligned}
\frac{\mathrm{d}}{\mathrm{d}x} 2^x &= \frac{\mathrm{d}}{\mathrm{d}x} e^{x\ln(2)} \\
&= \frac{\mathrm{d}}{\mathrm{d}x} e^x \bigg|_{x = x\ln(2)} \cdot \left[\frac{\mathrm{d}}{\mathrm{d}x} (x \ln(2)) \right] \\
&= e^{x\ln(2)} \cdot \left[\frac{\mathrm{d}}{\mathrm{d}x} (x \ln(2)) \right]
\end{aligned}$$

Example: $f(x) = 2^x$

Use the product rule to compute $\frac{d}{dx}(x\ln(2))$

$$\frac{d}{dx}(x\ln(2)) = \cancelto{1}{\left(\frac{d}{dx}x\right)} \cdot \ln(2) + x\cancelto{0}{(\ln 2)}$$
$$= \ln(2)$$

Example: $f(x) = 2^x$

Replace and simplify the result

$$
\begin{aligned}
\frac{\mathrm{d}}{\mathrm{d}x} 2^x &= e^{x \ln(2)} \ln(2) \\
&= 2^x \ln(2)
\end{aligned}
$$

$$\boxed{f'(x) = 2^x \ln(2)}$$

Derivative of Exponential functions $f(x) = n^x$

Now, if we repeat the same process with $f(x) = n^x$ we get the following rule,

$$\boxed{\frac{\mathrm{d}}{\mathrm{d}x} n^x = n^x \ln(n)}$$

Class Activity

Compute the derivatives of the following functions,

$$f(x) = \left(\frac{1}{2}\right)^x, \qquad g(x) = 3\left(\frac{1}{2}\right)^{x/2}$$

$$f'(x) = -\frac{1}{2^x}\ln(2), \qquad g'(x) = -2^{-x/2-1}\ln(8)$$

Power rule: $\frac{\mathrm{d}}{\mathrm{d}x}x^n = nx^{n-1}$ Chain rule: $\frac{\mathrm{d}}{\mathrm{d}x}u(v(x)) = u'(v(x))\cdot v'(x)$

Quotient rule: $\frac{\mathrm{d}}{\mathrm{d}x}\frac{u(x)}{v(x)} = \frac{u'v - uv'}{v^2}$ Product rule: $\frac{\mathrm{d}}{\mathrm{d}x}u(x)v(x) = u'v + uv'$